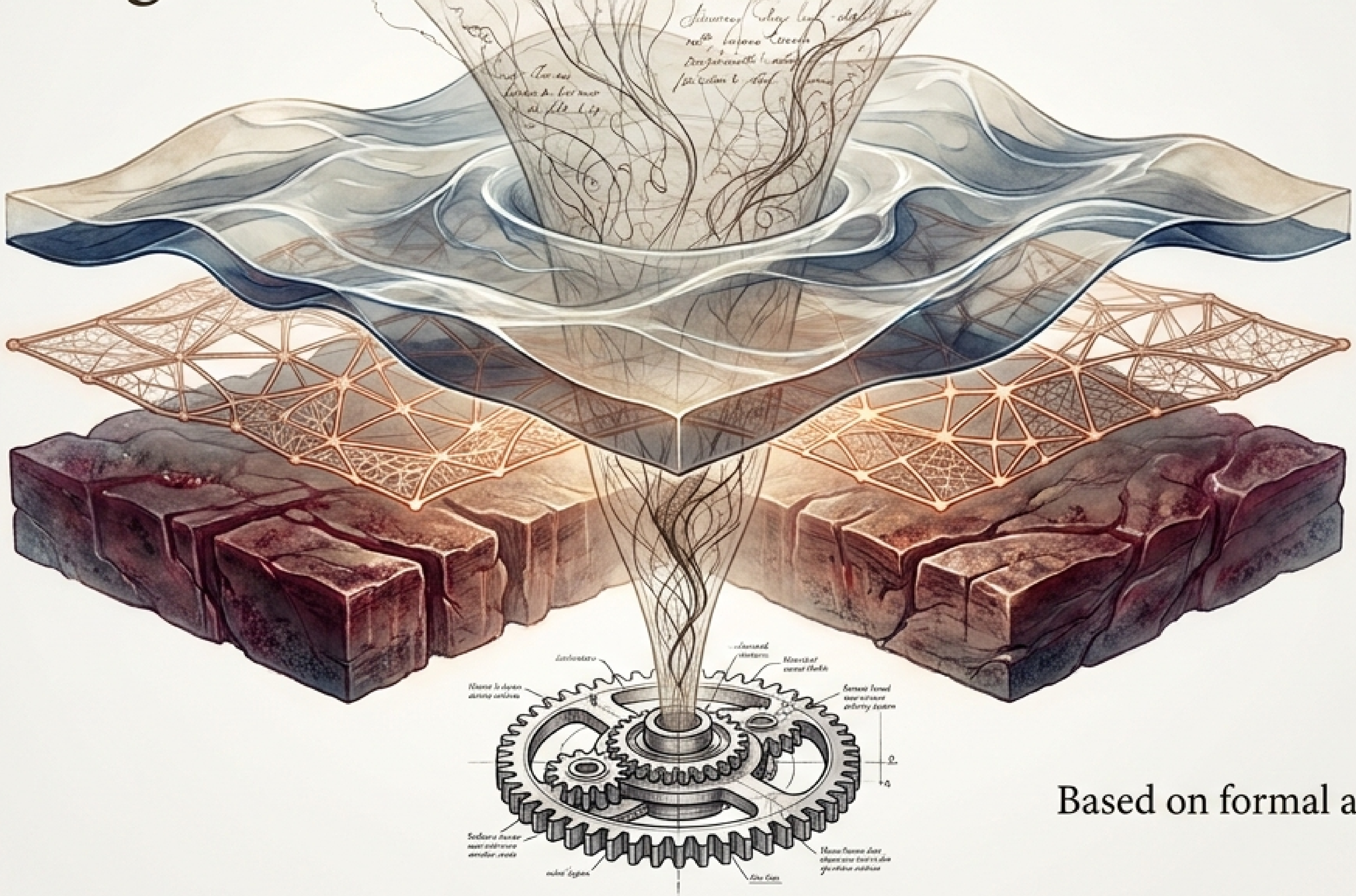
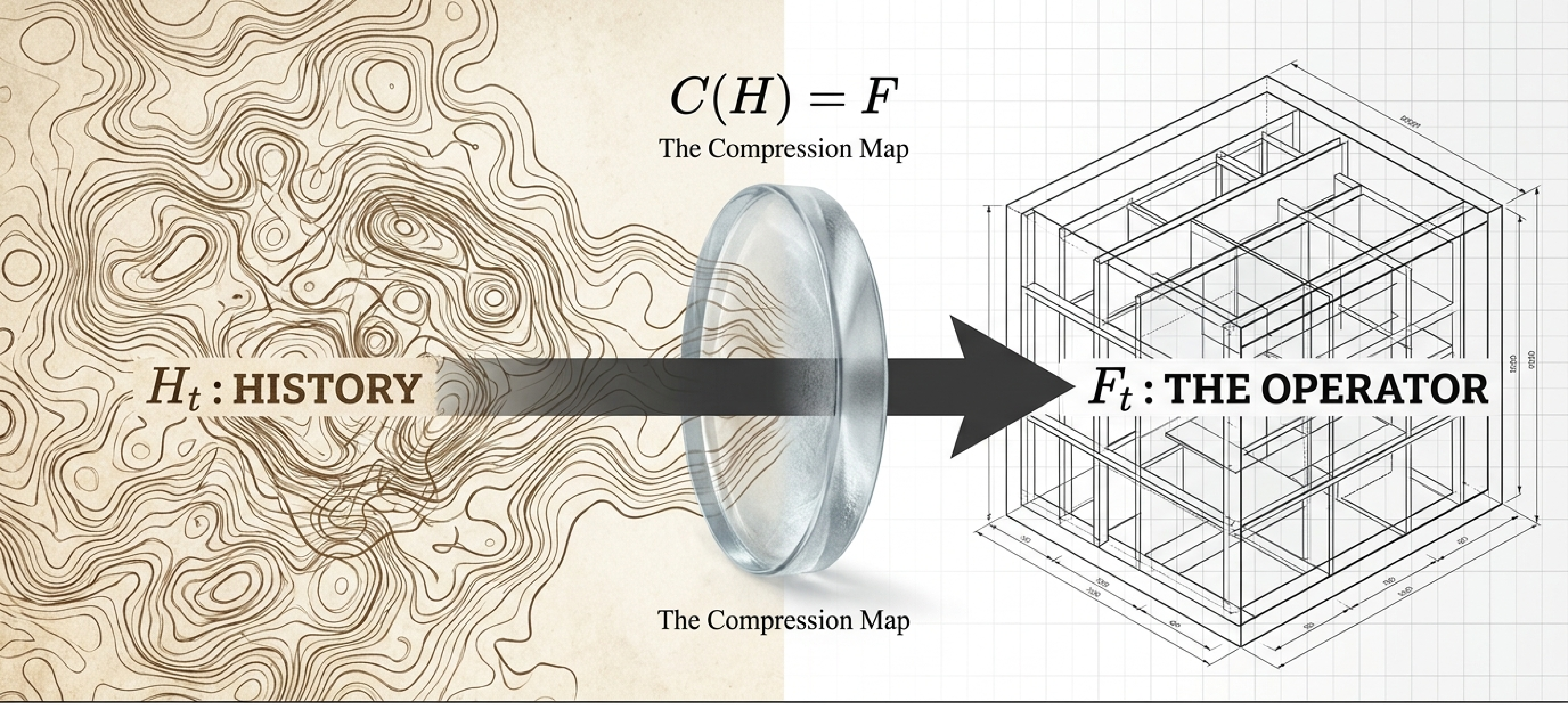
HISTORIES BECOME OPERATORS

Biological Notes on Retained Admissibility



Based on formal apparatus by Flyxion

July 2026



History Crystallized into Operative Constraint

The operator (F_t) governing a system's continuation is not a primitive object standing apart from its history. It is the compression of that history (H_t) into an operative constraint. The levels of a system are not primitive tiers standing beside one another. They are different depths at which history has been retained as a constraint on the present.

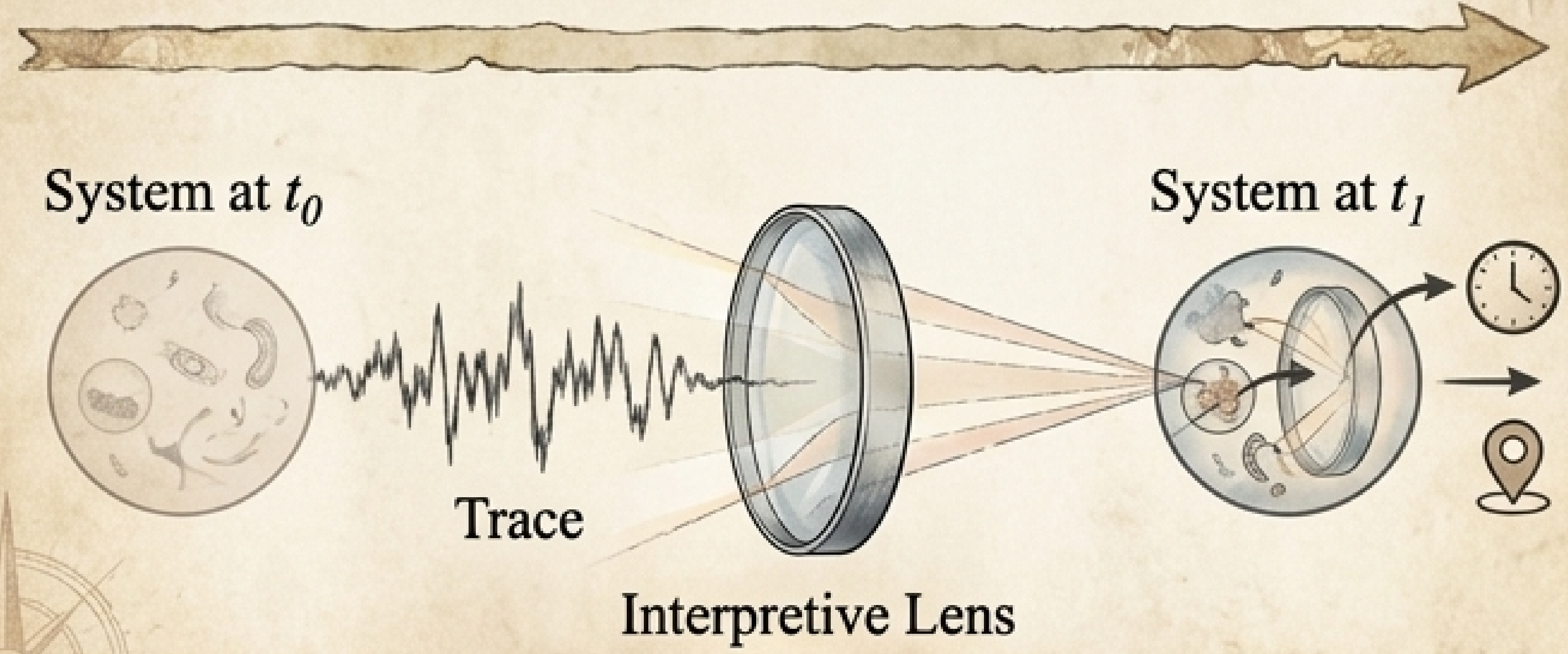
Separation I: Trace is Not Storage

	The Habit of Engineering	The Biological Premise
Goal	Fidelity	Creativity & Salience
Mechanism	Read / Write	Continuous Interpretation
Vulnerability	System Defect	Unreliable Substrate
Locus of Meaning	External Observer / Designer	Internal Agent

> *"A living system never has access to its own past; what it has are the memory traces that your past you has left... messages from your past self."* — Michael Levin

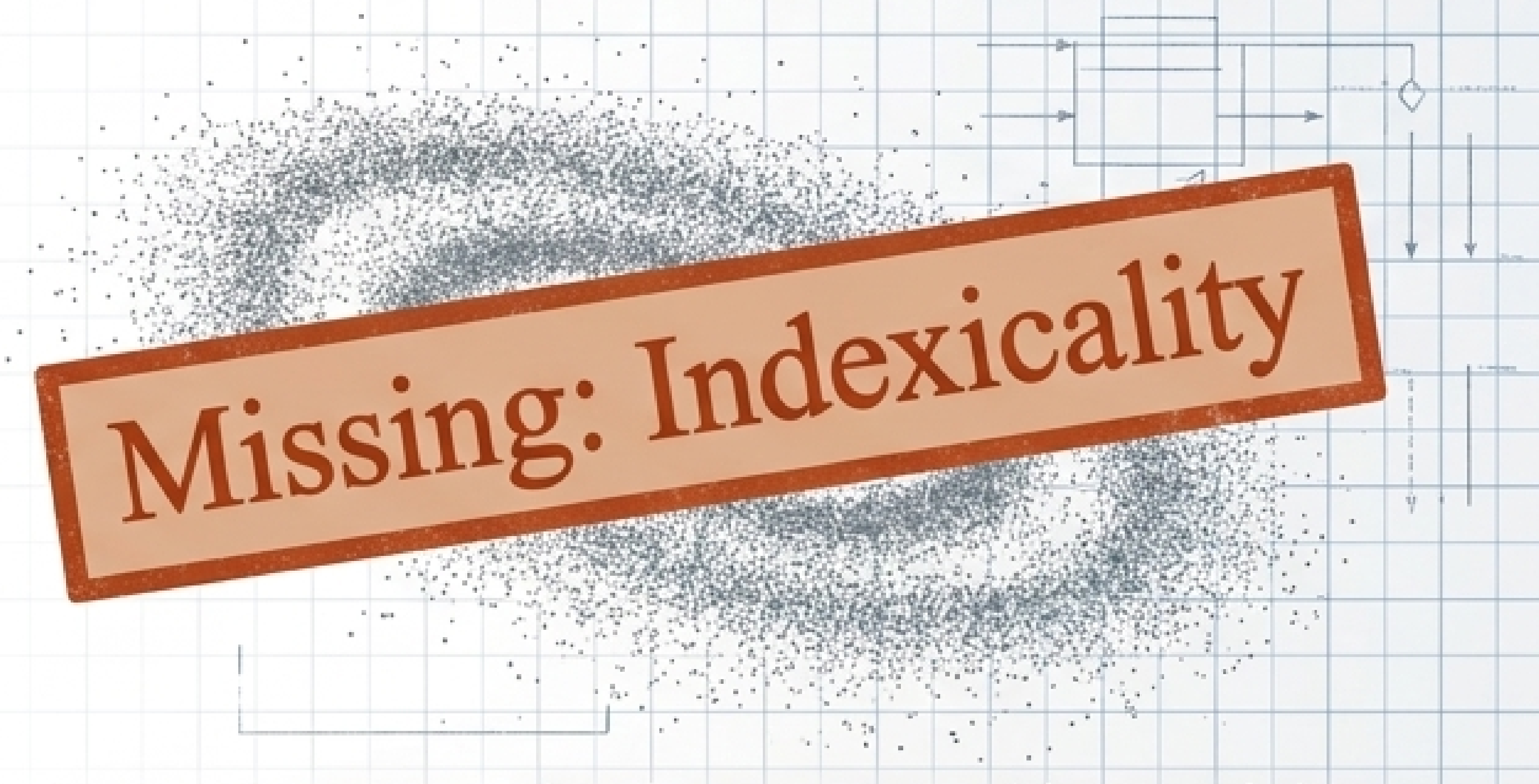
The Episodic Gap in Artificial Architectures

Episodic Memory (Biology)



Traces must be interpreted dynamically. Records what happened, when, and under what circumstances.

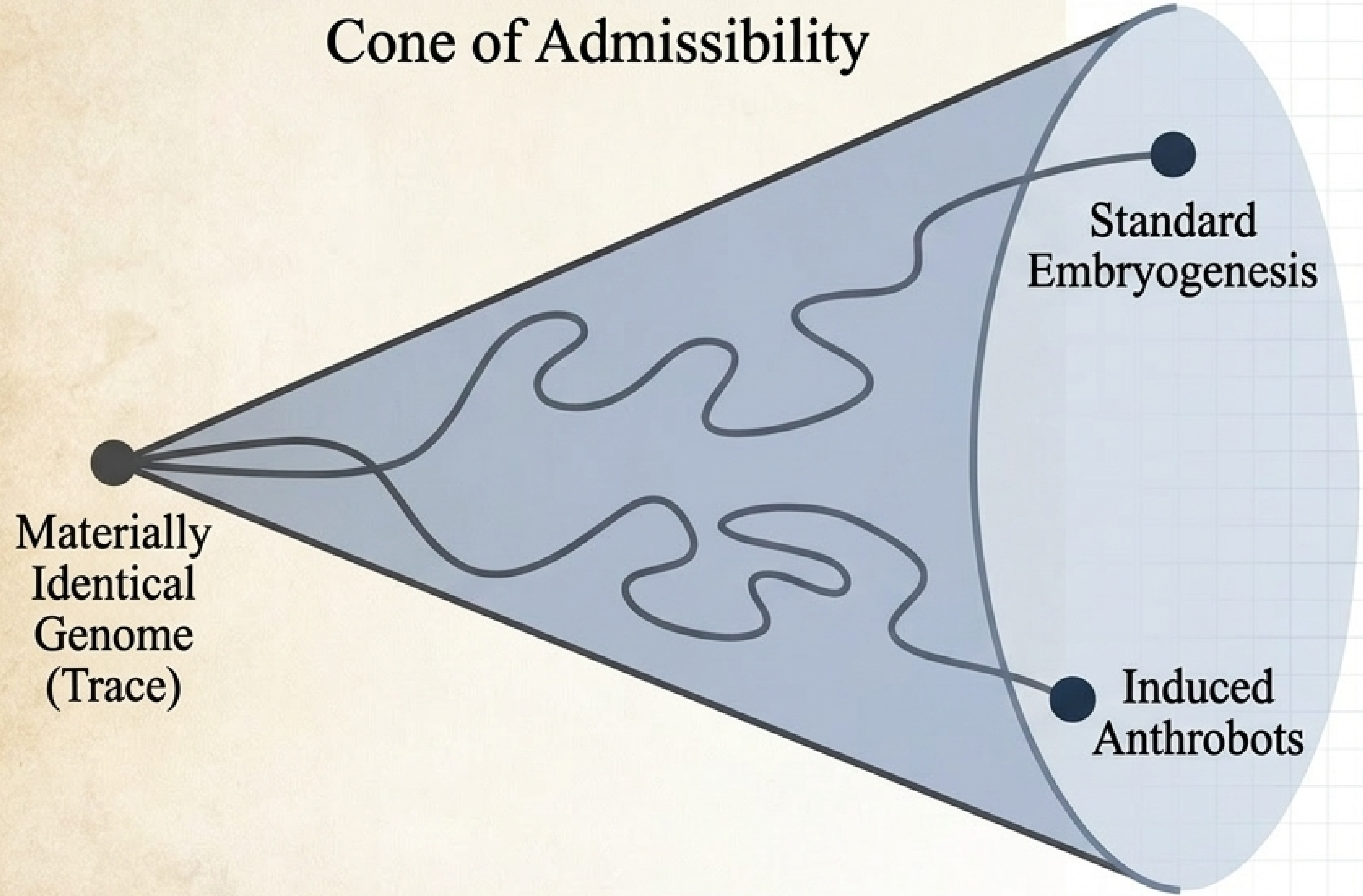
Pure Semantic Memory (Current LLMs)



The residue of training is compressed into weights that were never indexed to a particular episode.

Insight: Dumping conversational history into a retrieval store treats the symptom, not the structure. The missing ingredient isn’t storage fidelity—it’s memory tied to a specific past occasion.

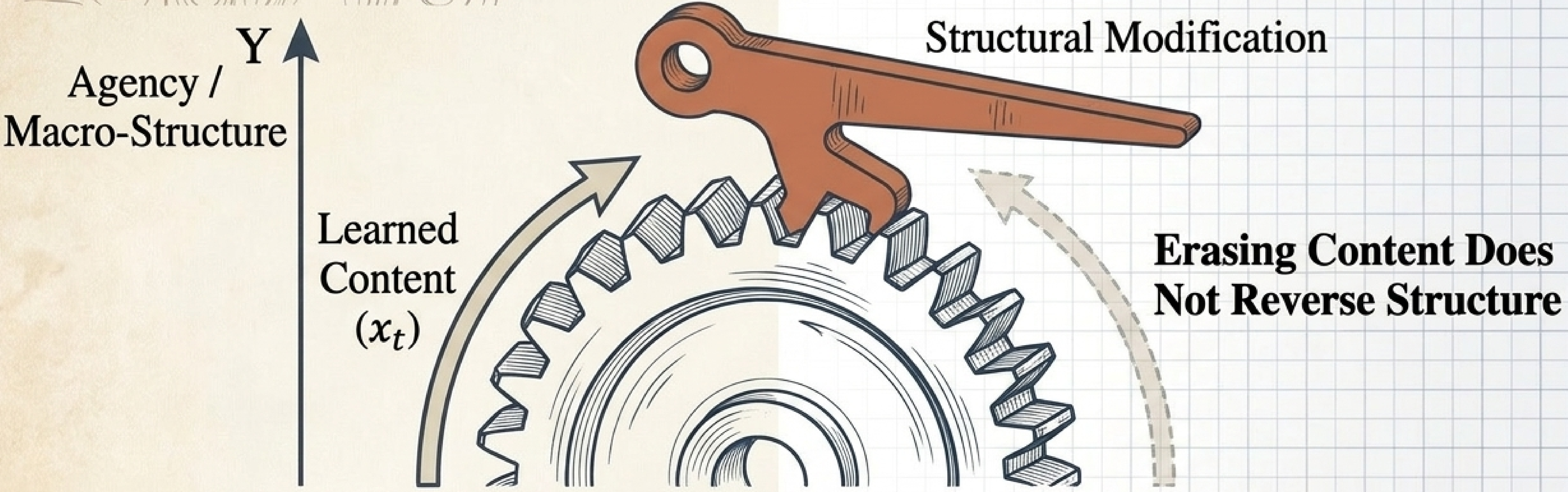
Separation II: Constraint is Not Determination



Core Concept: History functions as a constraint on admissible futures, not as a specification of one future in particular.

- The Genomic Evidence: The genome is not a blueprint; it is a heavily underdetermined score. The same genome diverges into radically different anatomies coherently.
- Takeaway: The trace rules out an enormous range of futures, but privileges no single correct reading among the remaining valid options.

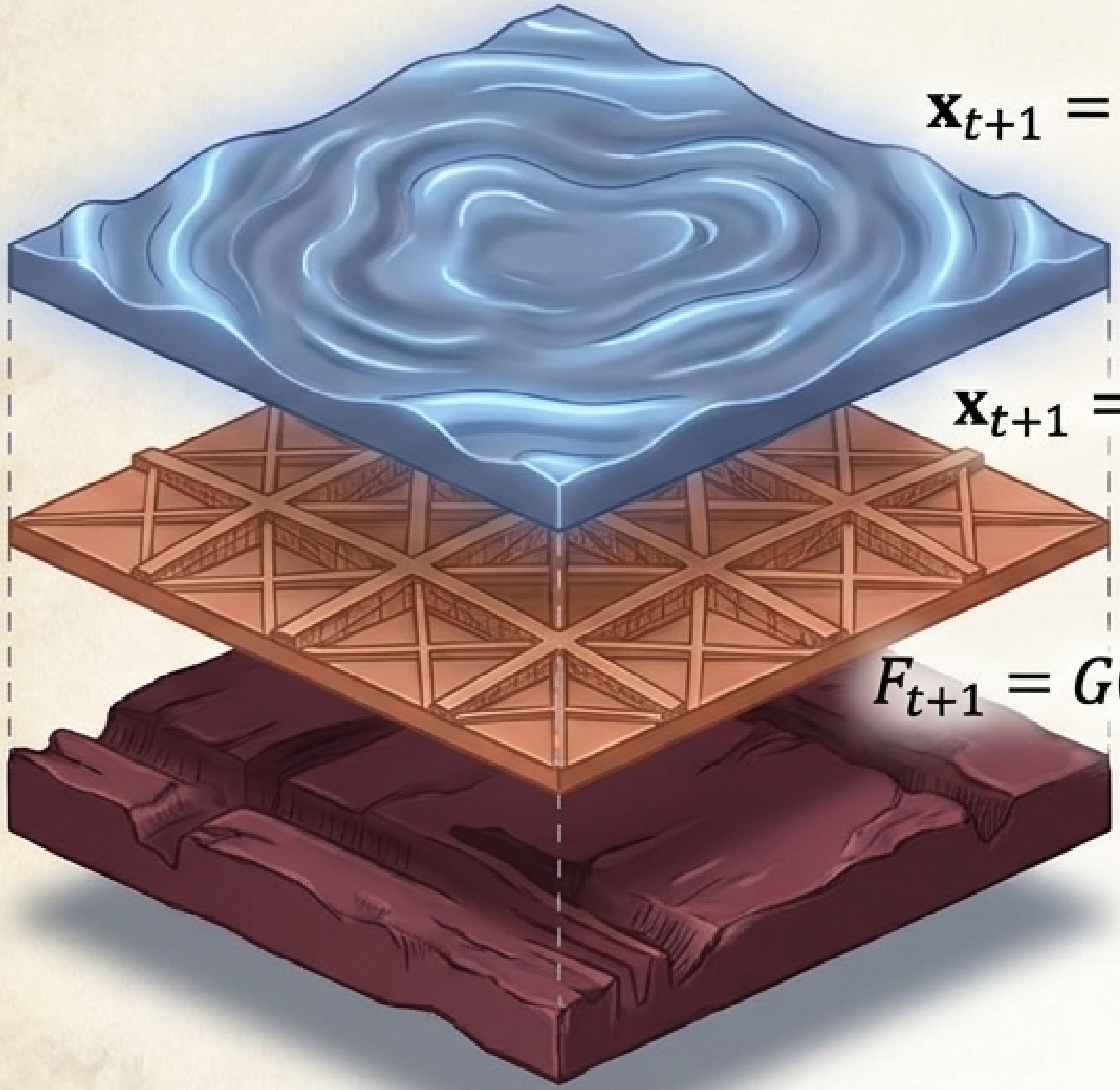
Separation III: Robustness is Not Control



The Pegozzi Result: A system built to interpret its own traces gains adaptive plasticity at the exact cost of specific specifiability.	The Mechanism: Systems with substantial causal causal emergence learn well. Training increases their causal emergence.	The Asymmetric Ratchet: Forgetting learned content does not reverse the gain in causal emergence. What survives is a structural change in the system's capacity.

The Adaptation Hierarchy

Depths of Historical Compression



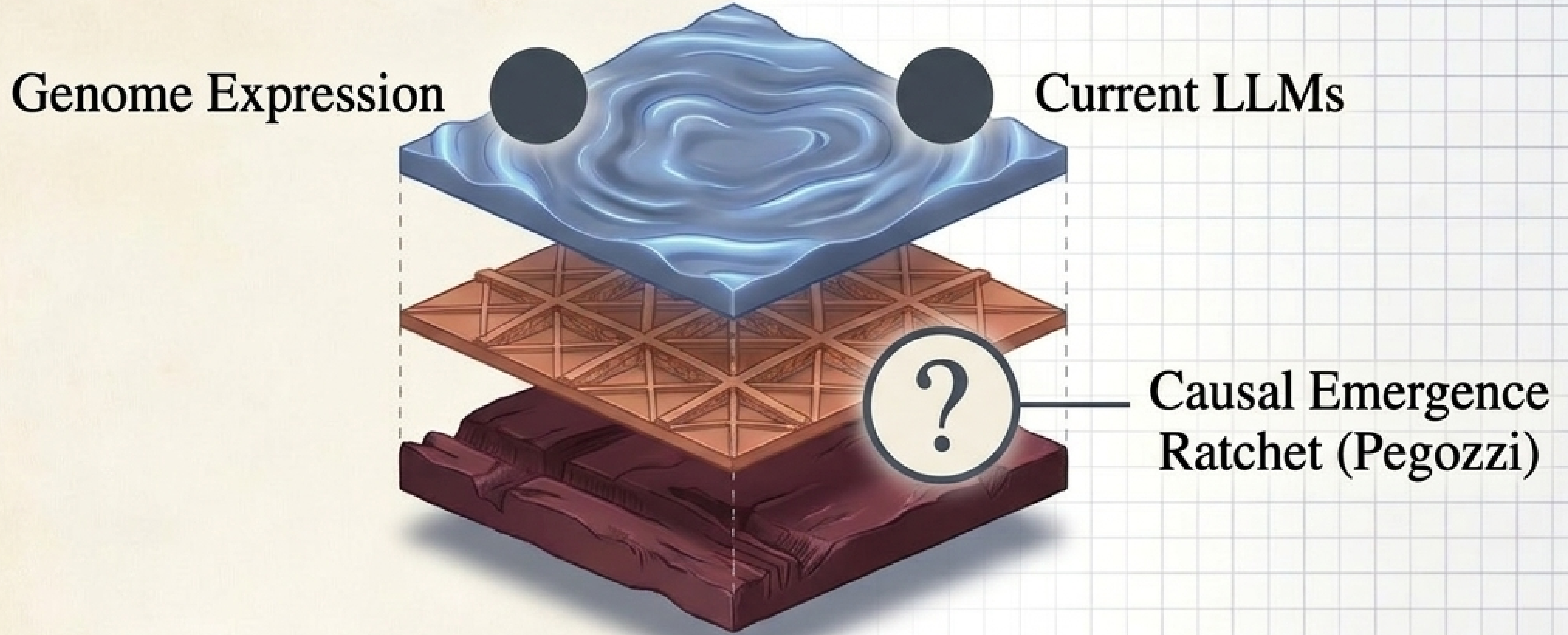
Definition 5.1 (Level 1: State Update) -
A fixed mechanism F acts on a changing state.

Definition 5.2 (Level 2: Parameter Update) -
The functional form F is fixed, but a parameter θ_t varies within it.

Definition 5.3 (Level 3: Rule Update) -
The mapping itself, not merely a parameter, is modified by process G.

Warning: Conflating Level 2 (parameter shifts) and and Level 3 (rule rewrites) is a categorical error.

Placing the Biological and Artificial Cases

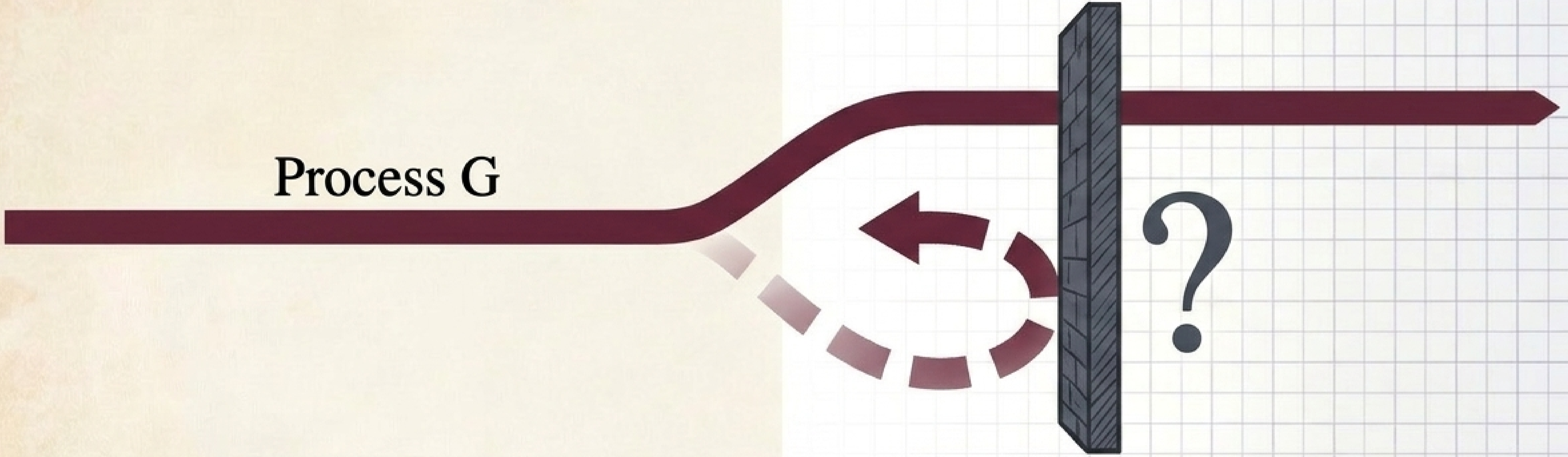


The Developmental Case (Level 1): The genome (**F**) remains fixed. Variability is in the outcome under changing environments, not in the rule itself.

The Episodic Gap (Level 1): Inference is a state update with no online parameter or rule revision. The revision boundary is placed entirely outside the episode.

The Causal Emergence Ratchet (Level 2 or 3): History is retained deeper than the state—as a change to the interpretive mechanism itself. The exact depth remains an open question.

The Open Question of Irreversibility



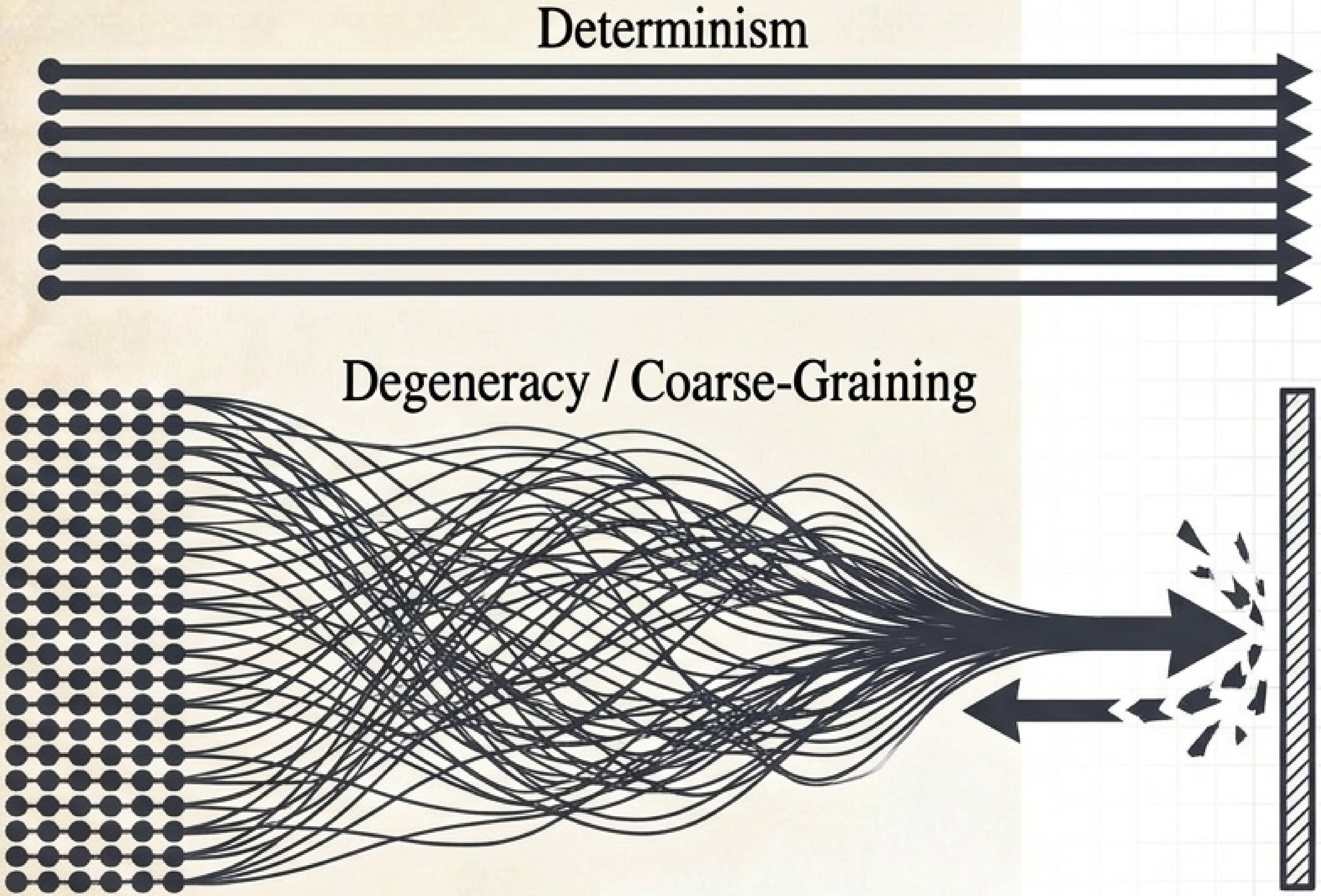
Question 6.1: Under what conditions does process G inherit irreversibility from H_t?

The Core Mystery: The causal-emergence result proves that learning-induced revision can be path-asymmetric (structural gain survives the erasure of state x_t).

The Frontier: Is this asymmetry a general mathematical property of Level 2, specific only to Level 3, or a further condition entirely? Neither biological observations nor archival proofs have yet resolved this at any depth of the hierarchy.

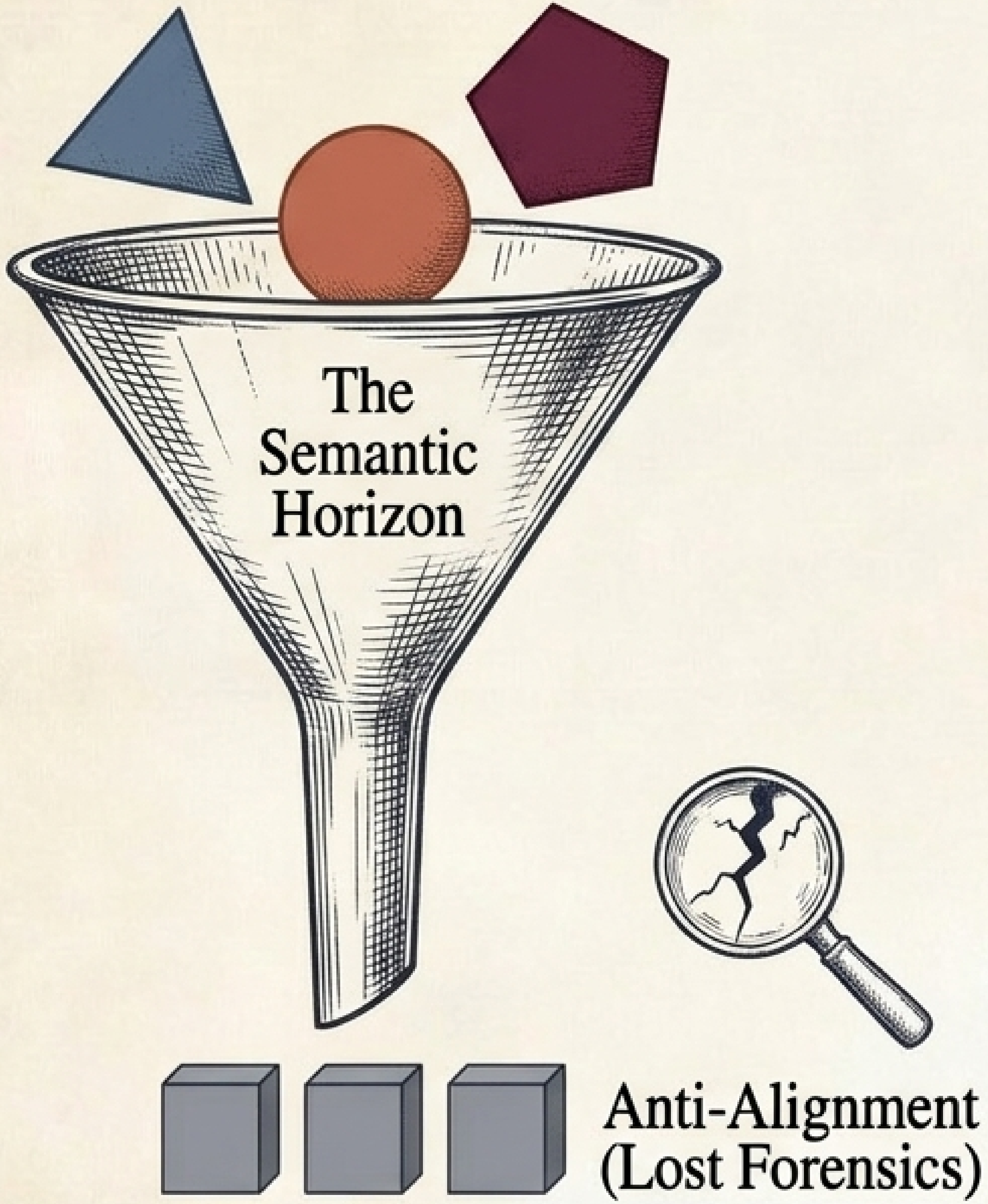
The Candidate Mechanism: Degeneracy Reduction

Causal Flow Diagrams



- **Effective Information** (Hoel): Decomposes as Determinism minus Degeneracy.
- **Conjecture 7.1:** Process G inherits irreversibility when the change induced in F_t is predominantly degeneracy-reducing.
- **The Thermodynamic Parallel:** Degeneracy reduction is structurally identical to coarse-graining. Distinguishable micro-trajectories collapse onto a shared macro-consequence. Inverting the collapse is generically non-unique because the compression removes the very information required to un-compress it.

The Cost of Compression

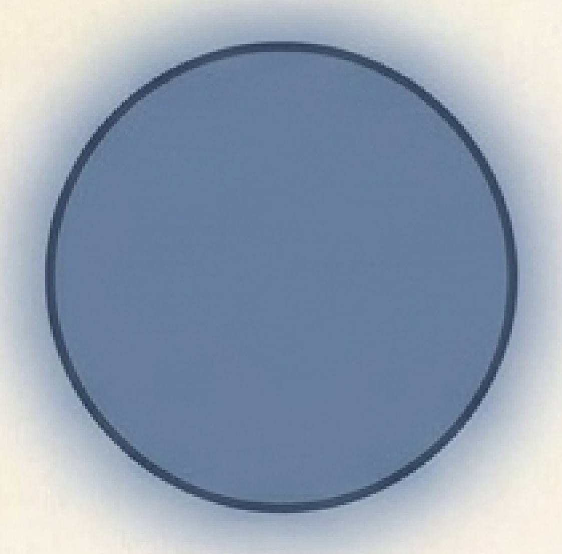


The Semantic Horizon: Recoverability falls as the inverse of preimage size. As distinct classes merge, causal influence persists, but recoverability drops to zero.

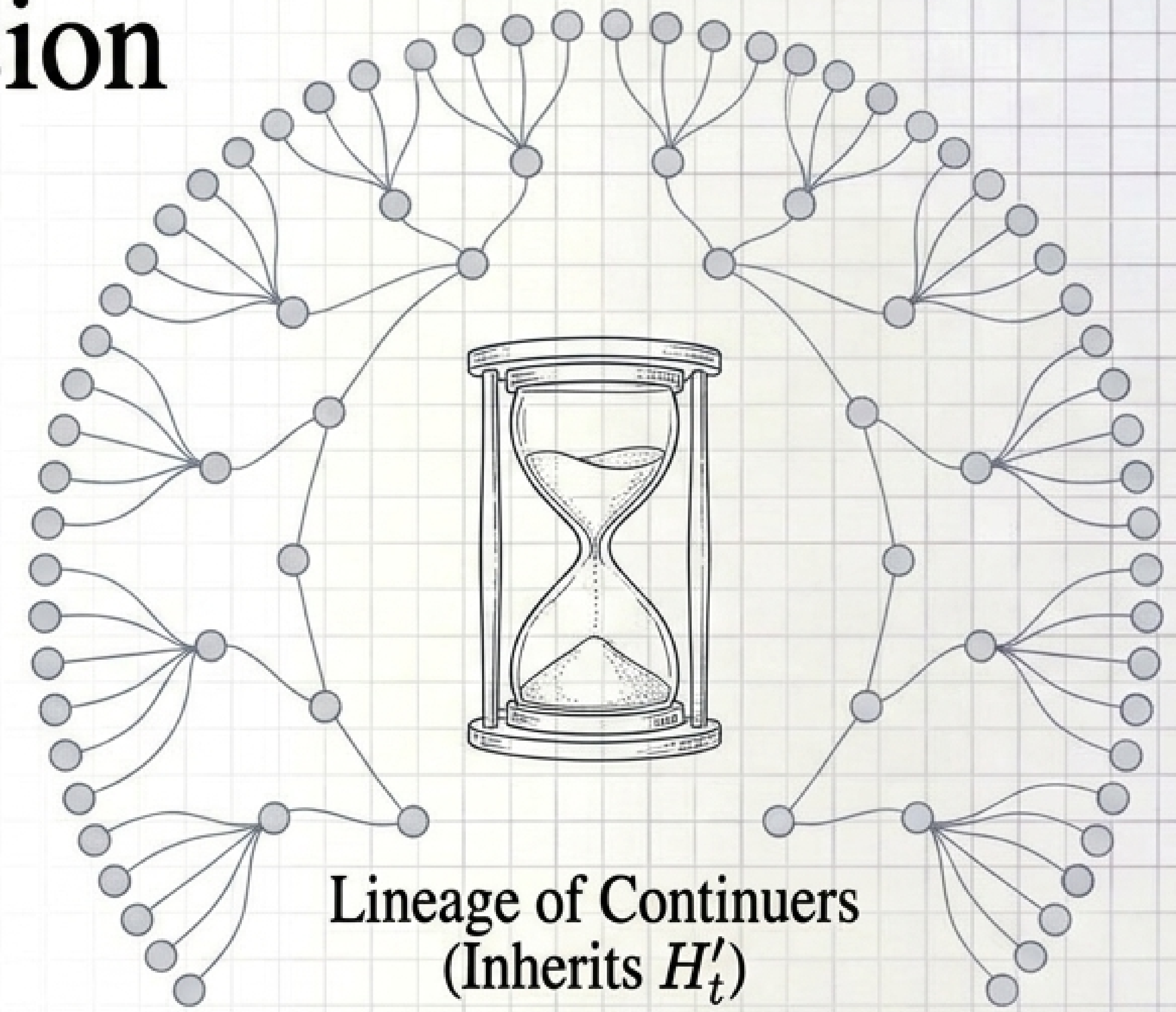
History Before Function (Anti-Alignment): Over the lattice of retained traces, the most minimal retention weakly maximizes repair entropy.

The Insight: What is disproportionately lost during compression is exactly the diagnostic detail needed to tell which prior configuration produced the current one. Erasing the present state (x_t) cannot restore that detail, because efficient compression never retained it.

A Coda: Operator Succession Across Continuers



Single Continuer
(Inherits H_t)



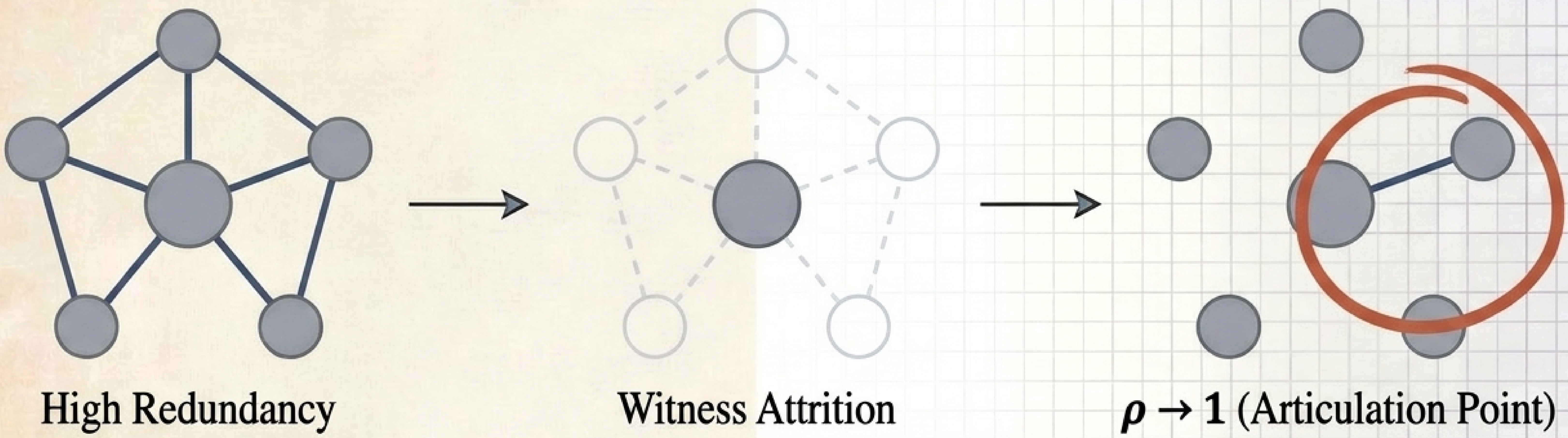
Lineage of Continuers
(Inherits H'_t)

The Shift in Scope: The previous cases answer how history is compressed within a single system. We now ask: what happens across a population when the direct carriers of history are mortal?

The Case of The Functional Melancholic: A generation holding direct, embodied memory of a pre-digital environment possesses comparative access. They ran the history themselves.

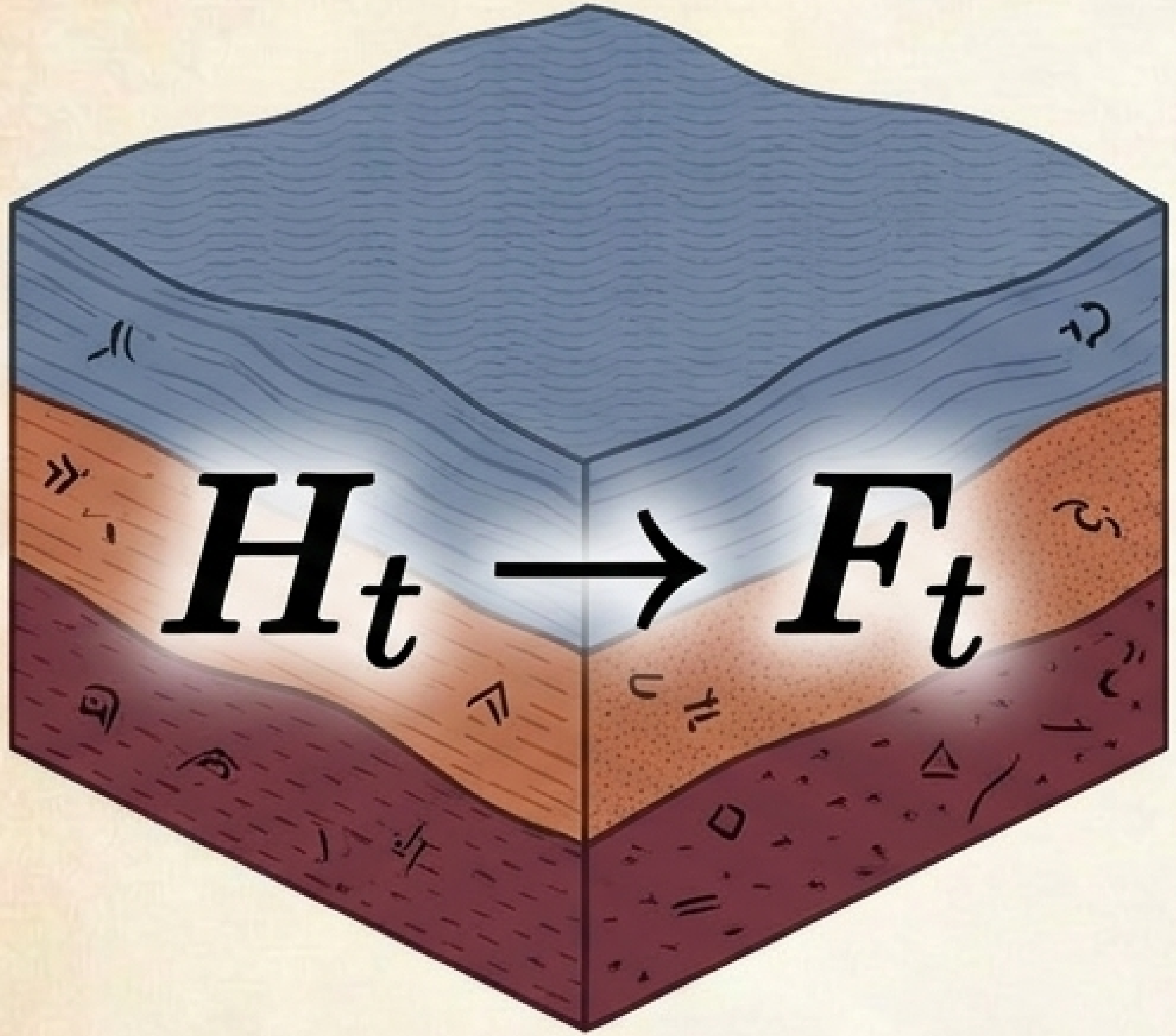
The Loss: Later generations inherit only H'_t — a compressed, already-interpreted account — with no outside vantage point from which to evaluate the current regime.

Monopoly in Reconstruction Redundancy



- **The Second-Order Repair Operator (W):** Maps histories to the repair capacities they can reconstruct.
- **Redundancy (ρ):** The number of independent historical pathways from which a capacity can be regenerated.
- **The Articulation Point:** When a generation of firsthand witnesses approaches extinction, repair capacity approaches $\rho(r) = 1$.
- **Takeaway:** Once the final independent pathway is destroyed, recovery is replaced by mathematical irreversibility. This is not ordinary forgetting; it is the structural severing of a graph.

Memory is Retained Admissibility



1. **History is not Stored, it is Interpreted:** Biological memory optimizes for the creativity of the reading, not the fidelity of the trace.
2. **Constraint Drives Adaptive Agency:** The narrowing of admissible futures, operating through causal emergence, trades exact control for robust plasticity.
3. **Compression is Irreversible:** Whether within a single system collapsing micro-trajectories, or across a dying lineage losing its witnesses, the compression of history into operative constraint irrevocably destroys diagnostic detail.

An organism is its own interpreter.
Its mechanisms are the crystallized history of its continuation.